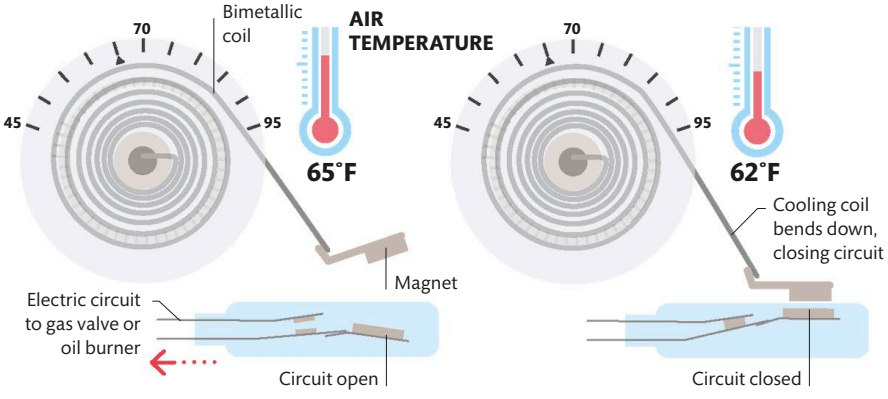
## Thermostats

Used to maintain temperatures in a home, thermostats may be local—fitted to a room—or global. When the temperature drops below the temperature set by the user, the thermostat completes a circuit that sends a signal instructing the boiler to fire and generate more heat.

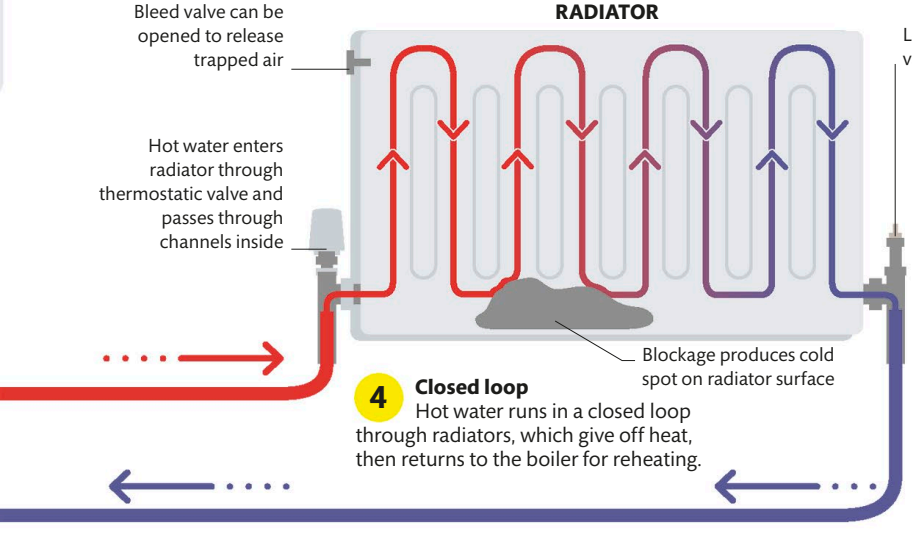



**1 Warm enough**
When the temperature is higher than the desired setting (64°F/18°C in this example), the coil warms up and straightens, pulling the magnet away from its contact and breaking the circuit. The boiler stops firing.

**2 Cooler than desired**
As it cools, the coil bends and the magnet moves toward the contact. The contact closes and completes the electric circuit, which sends a signal to the boiler to fire and heat water.

## Central heating

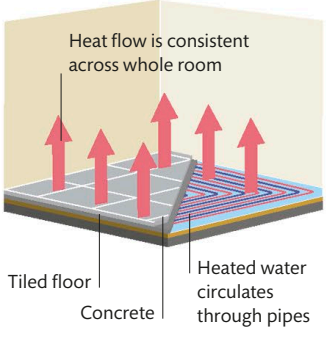
Hot water is pumped from the boiler through pipes or channels inside radiators, heating the radiator's outer panels, which warm the surrounding air. The lockshield valve adjusts the speed of the water's flow through the radiator—a slower flow makes radiators become hotter.




**4 Closed loop**
Hot water runs in a closed loop through radiators, which give off heat, then returns to the boiler for reheating.

## UNDERFLOOR HEATING

There are two main types of underfloor heating systems. A wet system uses networks of pipes or tubes through which heated water is pumped. Dry systems feature heated coils powered by electricity. Both can be expensive to install and run, but they can let heat radiate up through the floor to warm an entire room evenly without cold spots.




**WATER-BASED (WET) UNDERFLOOR HEATING**

### DOES TURNING UP A THERMOSTAT HEAT A HOUSE FASTER?

No. When a thermostat is set, the boiler runs at its maximum until the desired temperature is reached. It does not operate faster to reach higher temperatures.